

Question:-

A neural network predicts student scores with a Mean Squared Error (MSE) of 0.85 during training and 5.20 during testing. Analyzing the possible reasons for the difference and suggest suitable corrective measures.

Solution:-

Given data

$$\text{Training MSE} = 0.85$$

$$\text{Testing MSE} = 5.20$$

Calculate the difference in error.

Formula:-

$$\text{Difference in MSE} = \text{Testing MSE} - \text{Training MSE}$$

Calculation:-

$$= 5.20 - 0.85$$

$$= 4.35$$

Result:-

The difference between training and testing error is

4.35

Step 3:-

Error Analysis:-

Bias Over fitting:-

The neural network has learned the training data too closely, including its noise and patterns. As a result, it performs very well on training data but poorly on new (testing data).

- Training MSE = 0.85 (LOW)
- Testing MSE = 5.20 (High)

Reason 2: Insufficient Training Data

The model may have been trained using a small dataset. A limited amount of data makes it difficult for the network to learn general patterns.

Complex Neural Network:-

The neural network may contain too many hidden layers or neurons, making it memorize the training data instead of learning meaningful relationships.

Data Distribution Difference:-

The training and testing datasets may not have similar characteristics. If the testing data differs significantly, the model's prediction accuracy decreases.

Lack of Regularization:-

The Model May not use regularization. Methods such as Dropout or L2 regularization, increasing the risk of overfitting.

Step 4 (Corrective Measures)

1. Increase Training Data

Collect More student score data so that the Model learns a wider variety of patterns.

2. Apply Regularization:-

Use techniques such as:-

- Dropout

- L1 Regularization

- L2 Regularization

3. Simplify the Model:-

Reduce the no of hidden layers or neurons if the network is unnecessarily complex.

4. Use early stopping:-

stop training when the validation to evaluate errors.

starts increasing. This prevents the Model overfitting the training data.

5. perform cross-validation:-

Use k-fold cross validation to evaluate the Model on different subsets of the data and improve reliability.

output:

Interpretation of Results:

- The training MSE (0.85) is much smaller than the testing MSE (5.00).
- The MSE difference is 4.15, which is significant.
- This indicates that the model has overfitted to the training data.
- The model has low training error but poor generalization on unseen data.

Conclusion:-

The neural network showing overfitting because the training MSE (0.85) is much lower than the testing MSE (5.00). The large error difference (4.15) indicates poor generalization. By using Regularization, increasing training data, simplifying the network, early stopping, cross-validation, and proper data preprocessing, the model can achieve better performance on unseen data.

Key